第八章 电磁感应

动生电动势：

$\vec f = e\vec v\times\vec B$

$\vec f + e\vec E = 0$

$\therefore \vec E = -\vec v\times\vec B$

$\varepsilon = \int(\vec v\times\vec B)\cdot d\vec l$

感生：

$\oint \vec E_{感}\cdot d\vec l = -\iint \frac{\partial \vec B}{\partial t}\cdot d\vec S$

$\rightarrow \nabla\times\vec E_{感} = -\frac{\partial\vec B}{\partial t}$

而 $\nabla\times\vec E_{静} = 0$

$\therefore \nabla\times\vec E = \nabla\times(\vec E_{感}+\vec E_{静}) = -\frac{\partial \vec B}{\partial t} \neq 0$

⇒ 普遍情况下电场不是保守场

线：$-\frac{d}{dt}\iint \vec B\cdot d\vec S = -\frac{\partial}{\partial t}\int \vec A\cdot d\vec l = -\int \frac{\partial\vec A}{\partial t}\cdot d\vec l$

$= \oint \vec E_{感}\cdot d\vec l$

$\rightarrow \vec E_{感} = -\frac{\partial\vec A}{\partial t}$

$\vec E = \vec E_{感} + \vec E_{静} = -\nabla u - \frac{\partial \vec A}{\partial t}$

一般形式，$\varepsilon = -\frac{d}{dt}\iint\vec B\cdot d\vec S = -\iint\frac{\partial\vec B}{\partial t}\cdot d\vec S + \int(\vec v\times\vec B)\cdot d\vec l$

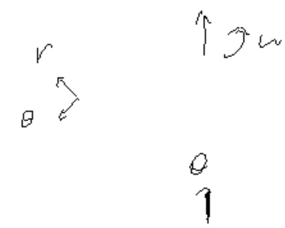
由于导体运动 f_1 ⟺ 导体内电场

由于电子漂移速度 f_2 ⟺ 外力（与安培力平衡）

$P = P_1+P_2$

$= (evB)\cdot u + (euB)(-v) = 0$ 洛伦兹力不作功

(b) f 1.07



$$B(\theta) = \frac{\mu_0}{4\pi}\left[\frac{3(\vec{m}_1\cdot\vec{r})\vec{r}}{r^5} - \frac{\vec{m}_1}{r^3}\right]$$

$$= \frac{\mu_0}{4\pi R^3}\left[3 m_1 \cos\theta\,\hat{r} - (m_1\cos\theta\,\hat{r} - m_1\sin\theta\,\hat{\theta})\right]$$

$$= \frac{\mu_0 m_1}{4\pi R^3}(2\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta})$$

$$d\varepsilon = \vec{v}\,dl \times \vec{B} = \omega R\sin\theta \cdot R\,d\theta \cdot \frac{\mu_0 m_1}{2\pi R^3}\cos\theta$$

$$\varepsilon_{tot} = \int_0^{\frac{\pi}{2}} d\varepsilon$$

$$= \frac{\mu_0\omega m_1}{2\pi R}\int_0^{\frac{\pi}{2}}\sin\theta\cos\theta\,d\theta = \frac{\mu_0\omega m_1}{4\pi R}$$

$$U_{AB} = 0$$

自感：

$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I\, d\vec{l} \times \vec{r}}{r^3} \Rightarrow B \propto I$

$\hookrightarrow \phi = \iint \vec{B} \cdot d\vec{S} \propto I$

$= L I$

$\varepsilon = -\frac{d\phi}{dt} = -L \cdot \frac{dI}{dt}$ 　 L：自感系数 ~ 几何形状 & 磁介质

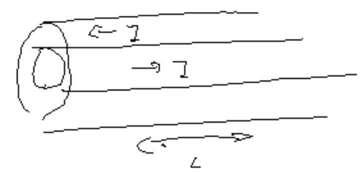
例1 p280 螺线环的自感系数.

$B = \mu_r \mu_0 n I$

$\psi = N \cdot \phi = N \cdot \mu_r \mu_0 n I \cdot S$

$L = \frac{\psi}{I} = \frac{N^2}{l} S \mu_r \mu_0 = n^2 \mu_r \mu_0 V$

例2 p321 同轴电缆单位长度的自感系数



$B = \frac{\mu_r \mu_0 I}{2\pi r}$

$\phi = \int_{R_1}^{R_2} B\, l\, dr = \frac{\mu_r \mu_0 I l}{2\pi} \ln\left(\frac{R_2}{R_1}\right)$

$\frac{\phi}{Il} = \frac{\mu_r \mu_0}{2\pi} \ln\left(\frac{R_2}{R_1}\right)$

自感电路 (RL电路)

$\varepsilon - L\frac{dI}{dt} = 0$ 　 $\Rightarrow I = I_0(1 - e^{-\frac{R}{L}t})$

$I = \frac{\varepsilon}{R}$

互感：

1 2

$$\phi_1 = \iint (\vec{B}_{11} + \vec{B}_{12}) \cdot d\vec{S}_1$$

$$= \phi_{11} + \phi_{12}$$

$$\phi_2 = \phi_{22} + \phi_{21}$$

$$M_{12} = \frac{\phi_{12}}{I_2} \quad M_{21} = \frac{\phi_{21}}{I_1}$$

$M = M_{12} = M_{21}$ 待证。

$$\begin{cases} \mathcal{E}_1 = -\frac{d\phi_1}{dt} = -(L_1 \frac{dI_1}{dt} + M_{12} \frac{dI_2}{dt}) = \mathcal{E}_{11} + \mathcal{E}_{12} \\ \mathcal{E}_2 = -\frac{d\phi_2}{dt} = -(L_2 \frac{dI_2}{dt} + M_{21} \frac{dI_1}{dt}) = \mathcal{E}_{22} + \mathcal{E}_{21} \end{cases}$$

互感系数相等的证明：

$$dW = dW_1 + dW_2$$

$$= \mathcal{E}_1 i_1 dt + \mathcal{E}_2 i_2 dt$$

$$= L_1 i_1 di_1 + L_2 i_2 di_2 + M_{12} i_1 di_2 + M_{21} i_2 di_1$$

$$= L_1 i_1 di_1 + L_2 i_2 di_2 + M_{12}\, d(i_1 i_2) + (M_{21} - M_{12})\, i_2 di_1$$

$$W = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M_{12} I_1 I_2 + (M_{21} - M_{12}) \int_0^{I_2} i_1 di_2$$

功和过程无关，=0

$\Rightarrow M = M_{12} = M_{21}$ (若为铁磁质，则有
$M_{12} \neq M_{21}$)

其他证明？

(1) 用磁矢势证明：

$$\phi_{12} = \iint_{S_2} B_1 \cdot dS_2 = \int_{l_2} A_1 \cdot dl_2$$

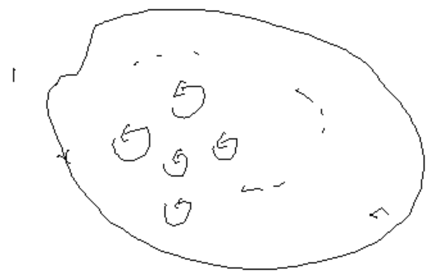
$$= \int_{l_2} \left(\int_{l_1} \frac{\mu_0}{4\pi} \frac{I_1 d\vec{l}_1}{r_{12}} \right) dl_2 = \frac{\mu_0 I_1}{4\pi} \oint_{l_2} \oint_{l_1} \frac{d\vec{l}_1 \cdot d\vec{l}_2}{r_{12}}$$

对称

$\Rightarrow M_{12} = M_{21}$

(2)

线圈 1 => 元分为小线圈，$d\vec{m} = I d\vec{S_1}$



$$\vec{B_{12}} = \frac{\mu_0}{4\pi} \iint_{S_1} \left(\frac{3(d\vec{m}\cdot\vec{r_{12}})}{r_{12}^5}\vec{r_{12}} - \frac{d\vec{m}}{r_{12}^3} \right)$$

$$= \frac{\mu_0 I_1}{4\pi r_{12}^3} \iint_{S_1} \left(3\,(d\vec{S_1}\cdot\vec{r_{12}})\,\vec{r_{12}} - d\vec{S_1} \right)$$

$$\phi_{12} = \iint_{S_2} \vec{B_{12}}\cdot d\vec{S_2}$$

$$= \frac{\mu_0 I_1}{4\pi r_{12}^3} \iint_{S_2}\iint_{S_1} \left[3\,(d\vec{S_1}\cdot\vec{r_{12}})(d\vec{S_2}\cdot\vec{r_{12}}) - d\vec{S_1}\cdot d\vec{S_2} \right]$$

对称

$\Rightarrow M_{12} = M_{21}$

习 8.3.25

(1) $B = \dfrac{\mu_0 I N a^2}{2(a^2+x^2)^{3/2}}$ $\qquad \phi_{12} = BS$

$$M = M_{12} = \frac{\phi_{12}}{I} = \frac{\mu_0 N a^2 S}{2(a^2+x^2)^{3/2}}$$

$$p_m = V\cdot J = V\cdot \mu_0 M = \mu_0 m = \mu_0 i S$$

$$\phi_{21} = M\cdot i$$

$$= \frac{\mu_0 N a^2 S i}{2(a^2+x^2)^{3/2}} = \frac{p_m N a^2}{2(a^2+x^2)^{3/2}}$$

$$\varepsilon = -\frac{d\phi}{dt} = \frac{p_m N a^2}{2}\cdot\frac{3}{2}\cdot\frac{2x\cdot v}{(a^2+x^2)^{5/2}}$$

$$= \frac{3 p_m N a^2 x v}{2(a^2+x^2)^{5/2}}$$

(2) $\phi_{21}(x) = \dfrac{p_m N a^2}{2(a^2+x^2)^{3/2}}$

$$Q = \int_0^{+\infty} I\,dt = \frac{1}{R}\int_0^{+\infty}\varepsilon\,dt = \frac{1}{R}\int_0^{+\infty} -\frac{d\phi}{dt}\,dt = \frac{1}{R}(\phi(0) - \phi(\infty))$$

$$= \frac{N p_m}{2a}$$

例 8.3.28.

$$H(r) = \frac{\Sigma I_0}{2\pi r} = \begin{cases} \frac{Ir}{2\pi R^2} & r<a \\ \frac{I}{2\pi r} & a<r<b \end{cases} \qquad B(r) = \begin{cases} \frac{\mu_1 Ir}{2\pi R^2} & r<a \\ \frac{\mu_2 I}{2\pi r} & a<r<b \end{cases}$$

① $\psi = \int_0^a \frac{\mu_1 Ir}{2\pi R^2} l\,dr + \int_0^b \frac{\mu_2 I}{2\pi r} l\,dr$

$= \frac{Il}{2\pi}\left(\frac{\mu_1 a^2}{2R^2} + \mu_2 \ln\left(\frac{b}{a}\right)\right)$

$\psi_e = \phi_e = \frac{\mu_0 Il}{2\pi} \ln\left(\frac{b}{a}\right) \qquad L_e = \frac{\psi_e}{I} = \frac{\mu_0 l}{2\pi} \ln\left(\frac{b}{a}\right)$

$d\psi_i = \frac{I_i}{I} d\phi_i = \frac{r^2}{a^2} \frac{\mu_1 I r}{2\pi R^2} l\,dr = \frac{\mu_1 I L}{2\pi a^2 R^2} r^3 dr$

$\psi_i = \int_0^a d\psi_i = \frac{\mu_1 I l}{8\pi a^2 R^2} \qquad L_i = \frac{\psi_i}{I} = \frac{\mu_1 l}{8\pi a^2 R^2}$

$L = \frac{\psi}{I} = \frac{\psi_i + \psi_e}{I} = \frac{\mu_1 l}{8\pi a^2 R^2} + \frac{\mu_2 l}{2\pi} \ln\left(\frac{b}{a}\right)$

$\frac{L}{l} = \frac{\mu_1}{8\pi a^2 R^2} + \frac{\mu_2}{2\pi} \ln\left(\frac{b}{a}\right)$

②

$dI = \frac{I}{\pi a^2} 2\pi r\,dr = \frac{2r\,dr}{a^2} I$

对于 $r' > r$ 处：$dH = \frac{dI}{2\pi r'}$，$dB = \mu_1 dH = \frac{\mu_1 dI}{2\pi r'}$

$d\phi_i = \int_r^a dB \cdot l\,dr' = \frac{\mu_1 l\,dI}{2\pi} \int_r^a \frac{dr'}{r'} = \frac{\mu_1 l\,dI}{2\pi} \ln\left(\frac{a}{r}\right)$

$= \frac{\mu_1 l I}{\pi a^2} r \ln\left(\frac{a}{r}\right) dr$

$\phi_i = \int_0^a d\phi = \frac{\mu_1 l I}{\pi a^2} \left(\int_0^a r \ln a\,dr - \int_0^a r \ln r\,dr\right)$

$= \frac{\mu_1 l I}{\pi a^2} \left(\ln a \cdot \frac{1}{2} a^2 - \frac{1}{2} r^2 \ln r \Big|_0^a + \frac{1}{2}\int_0^a r^2 \cdot \frac{dr}{r}\right)$

$= \frac{\mu_1 l I}{4\pi} \rightarrow L_i = \frac{\phi_i}{I} = \frac{\mu_1 l}{4\pi}$

为什么系数不对?

例 8.3.7

$B_1 = \mu_0 n_1 I_1 = \frac{\mu_0 N_1 I_1}{l}$, $B_2 = \frac{\mu_0 N_2 I_2}{l}$

$\phi_{11} = B_1 \pi a_1^2$, $L_1 = \frac{\Phi_{11}}{I_1} = \frac{\mu_0 \pi a_1^2 N_1}{l}$, $L_2 = \frac{\mu_0 \pi a_2^2 N_2}{l}$

$\phi_{12} = B_1 \pi a_1^2$, $M_{12} = \frac{\Psi_{12}}{I_1} = \frac{\mu_0 \pi a_1^2 N_1}{l} \cdot N_2$

$\phi_{21} = B_2 \pi a_1^2$, $M_{21} = \frac{\Phi_{21}}{I_2} = \frac{\mu_0 \pi a_1^2 N_2}{l} \cdot N_1$

电感串联问题

$$\varepsilon = \varepsilon_1 + \varepsilon_2$$

$$= \begin{cases} -L_1 \frac{dI}{dt} - L_2 \frac{dI}{dt} - M \frac{dI}{dt} - M \frac{dI}{dt} & b \to a' \\ -L_1 \frac{dI}{dt} - L_2 \frac{dI}{dt} + M \frac{dI}{dt} + M \frac{dI}{dt} & b \to b' \end{cases}$$

a b a' b'

$$\Rightarrow L = \begin{cases} L_1 + L_2 + 2M & \text{正接} \\ L_1 + L_2 - 2M & \text{反接} \end{cases}$$

a b a' b'

磁能

对于电路，

$dW = \varepsilon i\, dt = L i\, dt$

$W = \frac{1}{2} L I^2$

对于螺绕环，

$\phi = n l B S$ （l 为周长）

$\varepsilon = n l S \frac{dB}{dt} + iR$

安培环路定理，

$\oint H \cdot dl = \sum i$ $Hl = nil$

$\varepsilon i\, dt = i^2 R\, dt + n l S\, i \frac{dB}{dt} \cdot dt$

$= i^2 R\, dt + l S H\, dB$

$= \varepsilon I\, dt + V H dB$

$\Rightarrow$ 磁能密度 $w_B = H\, dB$

$\mu_r = \mu_r(\vec{H})$

$= \frac{1}{2} \mu_r \mu_0\, d(H^2)$ / $\mu_0 \cdot \vec{H} \cdot d(\mu_r \vec{H})$

利用 $W = \frac{1}{2} L I^2 = \frac{1}{2} H B$ 可求 L

互感：

$\varepsilon_{12} = -L \frac{dI_1}{dt} - M \frac{dI_2}{dt}$, $\varepsilon_{2} = -L_2 \frac{dI_2}{dt} - M \frac{dI_1}{dt}$

抵抗电动势做功

$dA = -\varepsilon_{12} I_1\, dt - \varepsilon_{21} I_2\, dt$

$= d(\frac{1}{2} L_1 I_1^2) - d(\frac{1}{2} L_2 I_2^2) - M I_1 dI_2 - M I_2 dI_1$

$= d(\frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M I_1 I_2)$

$\Rightarrow$ 互感能 $W_{mH} = M I_1 I_2$

离散情况下：

$W_m = \frac{1}{2} [\sum_{i=1}^{n} L_i I_i^2 + \frac{1}{2} \sum_{i \neq j} M_{ij} I_i I_j] = \frac{1}{2} \sum_{i=1}^{n} I_i \phi_i$

例 2 p 238.

$$H(r) = \begin{cases} \frac{\mu_0 I r}{2\pi R^2} & r < r_0 \\ \frac{\mu_0 I}{2\pi r} & r_0 < r < r_1 \\ \frac{\mu_0 I (r_2^2 - r^2)}{2\pi r (r_2^2 - r_1^2)} & r_1 < r < r_2 \end{cases}$$

$W_m = \frac{1}{2} \mu_0 \int H^2\, dV$

$= \frac{1}{2} [\int_0^{r_0} (\frac{\mu_0 I r}{2\pi R^2})^2 2\pi r l\, dr + \int_{r_0}^{r_1} (\frac{\mu_0 I}{2\pi r})^2 2\pi r l\, dr + \int_{r_1}^{r_2} (\frac{\mu_0 I (r_2^2 - r^2)}{2\pi r (r_2^2 - r_1^2)})^2 2\pi r l\, dr]$

...

例 8.4.3.

$L I + \phi = 0$

→ $I = \frac{BS}{L}$

$A = -\Delta W_m = \frac{1}{2}L\left(\left(\frac{BS}{L}\right)^2 - 0\right) = \frac{B^2S^2}{2L}$

位移电流：

因前有 $\frac{\partial \vec{B}}{\partial t}$，但无 $\frac{\partial \vec{E}}{\partial t}$

问题：非稳恒电流

$\oint_L H\cdot dl = \iint_S \vec{j}\cdot d\vec{S}$ → $\iint \vec{j}\cdot d\vec{S} = Const$

但 $\iint_{S_1} \vec{j}\cdot d\vec{S} = I \neq \iint_{S_2} \vec{j}\cdot d\vec{S} = 0$

对于电容器表面：

$\oint_{S_表} \vec{j}\cdot d\vec{S} = -\frac{dq_0}{dt}$, $\oint_{S_表} \vec{D}\cdot d\vec{S} = q_0$

$\oint_S [\vec{j}\cdot d\vec{S} + \frac{\partial}{\partial t}(\vec{D}\cdot d\vec{S})]$

S固定，$\frac{\partial}{\partial t}(d\vec{S}) = 0$

$= \oint_{S} (\vec{j} + \underline{\frac{\partial \vec{D}}{\partial t}})\cdot d\vec{S}$ ← 位移电流密度.

注意此时是闭合面.

⇒ $\oint_{\partial S} H\cdot d\vec{l} = \iint_{S} (\vec{j} + \frac{\partial \vec{D}}{\partial t})\cdot d\vec{S}$

$D = \varepsilon_0 \vec{E} + \vec{P}$ $\quad \frac{\partial \vec{D}}{\partial t} = \varepsilon_0 \frac{\partial \vec{E}}{\partial t} + \frac{\partial \vec{P}}{\partial t}$

($\oint \vec{P}\cdot d\vec{S} = -q'$ $\quad \oint \frac{\partial \vec{P}}{\partial t}\cdot d\vec{S} = -\frac{dq'}{dt}$ 极化电流)

例1 P336

$E = \frac{Q}{\varepsilon_0 S}$ $\quad D = \varepsilon_0 E = \frac{Q}{S}$

$I_d = J_d\cdot S = \frac{\partial \vec{D}}{\partial t}\cdot S = \frac{1}{S}\frac{dQ}{dt}\cdot S = \frac{dQ}{dt}$

例2 P336

$B\cdot 2\pi r = \mu_0 I_0 \cdot \frac{r^2}{R^2}$

例6.

$$E = \frac{Q}{4\pi\varepsilon_0 r^2} \qquad D = \varepsilon_0 E = \frac{Q}{4\pi r^2}$$

$$\phi_D = D \cdot S = \frac{Q}{4\pi r^2} \cdot 2\pi r(r-x) = \frac{Q(r-x)}{2r}$$

$$= \frac{1}{2}Q\left(1 - \frac{x}{\sqrt{a^2+x^2}}\right)$$

$$\boxed{I_d = \frac{\partial \phi_D}{\partial t}} = -\frac{1}{2}Q \cdot \frac{\sqrt{a^2+x^2} - x\cdot\frac{x}{\sqrt{a^2+x^2}}}{a^2+x^2} \cdot (-v) = \frac{Qva^2}{2(a^2+x^2)^{3/2}}$$

$$H \cdot 2\pi a = I_d \Rightarrow H = \frac{Qva}{4\pi(a^2+x^2)^{3/2}}$$

$$B = \mu_0 H = \frac{\mu_0 Q v a}{4\pi(a^2+x^2)^{3/2}}$$

如果是 $I_d = S \cdot \frac{\partial D}{\partial t}$ 呢?

$$D = \frac{Q}{4\pi}\frac{1}{a^2+x^2} \qquad \frac{\partial D}{\partial t} = \frac{Q}{4\pi}\frac{-1}{(a^2+x^2)^2}2x(-v)$$

$$I_d = S \cdot \frac{\partial D}{\partial t}$$

$$= 2\pi\sqrt{a^2+x^2}\left(\sqrt{a^2+x^2} - x\right)\frac{Q}{2\pi}\frac{xv}{(a^2+x^2)^2}$$

$$= Qv\left(\frac{x}{a^2+x^2} - \frac{x^2}{(a^2+x^2)^{3/2}}\right) \quad \times$$

附加1:

(1) $\oint E \cdot dl = -\frac{d\phi}{dt} \rightarrow E = \begin{cases} \frac{r}{2}\dot{B}_0 & r<R \\ \frac{\pi R^2}{2r}\dot{B}_0 & R<r \end{cases}$

(2) $J = \sigma E$

(3) $\frac{dQ}{V} = \frac{I^2 R\, dt}{V} = \frac{J^2 S^2 \cdot \rho\frac{l}{S}\, dt}{Sl} = \frac{1}{\sigma}J^2\, dt$

$$P = \frac{dQ}{dt} = \frac{1}{\sigma}\int_0^R \left(\sigma \cdot \frac{r}{2}\dot{B}_0\right)^2 \cdot 2\pi r d \cdot dr + \frac{1}{\sigma}\int_R^a \left(\sigma \cdot \frac{\pi R^2}{2r}\dot{B}_0\right)^2 2\pi r d \cdot dr$$

$$= \frac{\sigma\pi\dot{B}_0^2 d}{2}\int_0^R r^3\, dr + \frac{1}{2}\pi\sigma\dot{B}_0^2 R^4 d\, \ln\left(\frac{a}{R}\right)$$

$$= \frac{1}{8}\, 4\pi B_0^2\, d\, R^4 \left(1 + 4\ln\left(\frac{a}{R}\right)\right)$$

附加 2

$$-F \cdot v = -2\pi F \frac{dr}{dt}$$

$$= \frac{\mathcal{E}^2}{R} = \frac{1}{R}\left(\frac{d\Phi}{dt}\right)^2 = \frac{B^2}{R}\left(\frac{d}{dt}\pi r^2\right)^2$$

$$= \frac{\pi^2 B^2}{R}\, 4r^2 \left(\frac{dr}{dt}\right)^2$$

$$\Rightarrow \frac{dr}{dt} = -\frac{FR}{2\pi B^2 r^2}$$

$$\int_{r_0}^{0} r^2\, dr = \int_0^{t_0} \frac{FR}{2\pi B^2}\, dt$$

$$\frac{1}{3} r_0^3 = \frac{FR}{2\pi B^2} t_0 \quad \Rightarrow \quad t_0 = \frac{2\pi B^2}{3FR} r_0^3$$